

Statistics: Mean, Median, Mode, and Range

USA Grade 6 Math
CCSS 6.SP.A.1-3, 6.SP.B.5

Name: _____

Date: _____

LEARNING TARGET

I can summarize a data set and explain what measures of center and variability reveal.

KEY STRATEGY + WORKED EXAMPLE

Order the data, compute mean and median, identify mode and range, then choose the statistic that best represents the distribution.

For 2,4,4,6,9: mean=5, median=4, mode=4, range=7.

A. TRY TOGETHER - USE THE STRATEGY

1. Data [4, 6, 7, 7, 8, 10]: mean = ____
2. Data [12, 15, 15, 18, 20]: mean = ____
3. Data [3, 5, 5, 5, 9, 15]: mean = ____
4. Data [22, 24, 25, 27, 32]: mean = ____

B. CORE PRACTICE - SHOW YOUR WORK

5. Data [8, 9, 11, 11, 12, 15]: mean = ____
6. Data [40, 42, 45, 48, 50]: mean = ____

7. Data [2, 4, 6, 8, 10]: mean = ____

8. Data [13, 13, 14, 17, 18]: mean = ____

9. Data [5, 7, 7, 9, 12]: mean = ____

10. Data [21, 24, 24, 27, 29, 31]: mean = ____

11. Data [4, 6, 7, 7, 8, 10]: mean = ____

12. Data [12, 15, 15, 18, 20]: mean = ____

C. INDEPENDENT FLUENCY - CALCULATE AND CHECK

13. Data [3, 5, 5, 5, 9, 15]: mean = ____
14. Data [22, 24, 25, 27, 32]: mean = ____
15. Data [8, 9, 11, 11, 12, 15]: mean = ____
16. Data [40, 42, 45, 48, 50]: mean = ____
17. Data [2, 4, 6, 8, 10]: mean = ____
18. Data [13, 13, 14, 17, 18]: mean = ____
19. Data [5, 7, 7, 9, 12]: mean = ____
20. Data [21, 24, 24, 27, 29, 31]: mean = ____

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D. APPLY THE SKILL

21. For 6,7,7,8,9,25, find mean and median. Which better represents a typical value?

22. Compare sets A: 5,5,5,5 and B: 2,4,6,8 using mean and variability.

23. Create a five-value set with mean 12, median 10, and one outlier.

E. REASON, CHECK AND CORRECT

24. Explain how one high outlier affects mean and median differently.

25. Correct: The range of 4,9,12,18 is $14-4=10$.

EXIT TICKET + CHALLENGE

26. Analyze 3,5,5,8,9,12: mean, median, mode, range, and best center.

Challenge: Conduct a small survey, display the data, and justify the most informative summary statistic.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.

27. Explain one strategy you used today.

28. Check one answer in a different way.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: _____

31. The most difficult part was: _____

Why was it difficult? _____

32. My next practice step is: _____

TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

- A. 1) 7, 2) 16, 3) 7, 4) 26
- B. 5) 11, 6) 45, 7) 6, 8) 15, 9) 8, 10) 26, 11) 7, 12) 16
- C. 13) 7, 14) 26, 15) 11, 16) 45, 17) 6, 18) 15, 19) 8, 20) 26
- D. 21) Mean= $10\frac{1}{3}$, median=7.5; median better because 25 is an outlier. 22) Both means=5; A has no spread, B has range 6. 23) Example 5,9,10,11,25.
- E. 24) A high outlier raises mean more than median. 25) Range= $18-4=14$.
- EXIT. 26) Mean=7, median=6.5, mode=5, range=9; justify center from context.
- Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.
- F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may calculate before ordering data or confuse range with a count. Range is maximum minus minimum.

TEACHER CHECK

- ☐ Uses an appropriate Grade 6 Math strategy.
- ☐ Shows or explains thinking clearly.
- ☐ Checks a response using evidence, a rule, or a second method.
- ☐ Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.